

MOLECULAR PHYLOGENETICS & PHYLOGENOMICS — MOCK EXAM 3

A total of 27 closed questions, each worth one point, with no penalty for incorrect answers.

01. Define the primary difference between character-based and distance-based phylogenetic reconstruction:

- A. Character methods compress alignment columns into a single percentage matrix, while distance methods analyze each position independently.
- B. Distance methods compute a pairwise matrix tracking total changes, while character methods evaluate each individual alignment site against explicit evolutionary models.
- C. Character methods operate exclusively on protein matrices, while distance methods handle nucleotide strings.
- D. Distance methods require rooted chronograms, while character methods build unrooted cladograms directly.

02. Which nucleotide substitution model modifier adds a specific parameter to capture the exact fraction of alignment sites that have zero probability of mutating over evolutionary time?

- A. Gamma Distribution (+ Γ)
- B. Invariable Sites (+I)
- C. All Rates Different (ARD)
- D. Equal Rates (ER)

03. Choose the INCORRECT statement regarding the Maximum Parsimony (MP) tree construction paradigm:

- A. It scores candidate topologies by minimizing the total number of required mutations.
- B. It is heavily prone to systematic errors like Long Branch Attraction under rapid evolutionary rates.
- C. It relies exclusively on informative sites where a state appears in at least two separate taxa.
- D. It incorporates complex time-reversible substitution matrices to calculate branch lengths.

04. What do 'out-paralogs' represent inside a tracked gene family phylogeny?

- A. Gene duplicates that originated after a designated speciation boundary within a single organism.
- B. Paralogous genes that arose from a duplication event that occurred in the common ancestor before the speciation of the study group.
- C. Orthologous sequences that share an identical biological function across sister taxa.
- D. Genes modified by horizontal transfer loops between microbes.

05. Which sequencing approach provides the absolute lowest cost per sample but is strictly restricted to population genomics or fine-scale phylogenies of closely related species?

- A. Whole-Genome Sequencing
- B. RADseq / GBS
- C. Targeted Transcriptomics
- D. Mitochondrial Genome Assembly

06. In Bayesian phylogenetic MCMC implementations, a/an _____ chain is responsible for accurately sampling the true target posterior distribution, while parallel _____ chains explore a flattened landscape to escape local optima.

07. The Likelihood Ratio Test (LRT) statistic used to validate branch significance is calculated using which formula?

- A. $2(L1 + L0)$
- B. $2(L1 - L0)$
- C. $(L1 - L0) / 2$
- D. $L1 / L0$

08. A monophyletic clade is uniquely diagnosed and defined by the presence of:

- A. Sympleisiomorphies.
- B. Synapomorphies.
- C. Homoplasies.
- D. Autapomorphies.

09. In an edge-proportional partition model used for multi-gene supermatrix concatenation analysis:

- A. Every single partition is forced to share completely identical absolute branch lengths.
- B. Each partition has its own completely independent unlinked tree topology and branch length set.
- C. Partitions share a single global tree topology and relative branch lengths, but each partition has a scaling factor applied to its branch lengths.
- D. All partitions are forced to run under an empirical Jukes-Cantor model format.

10. Which of the following describes an exponential distribution prior used for node calibration in divergence time analysis?

- A. It assigns equal probability across a wide range between hard minimum and maximum age bounds.
- B. It peaks sharply directly at the specified minimum age constraint and declines rapidly into the past.
- C. It forms a symmetric bell curve centered around an optimized historical mean value.
- D. It incorporates a heavy asymmetric tail that reaches infinitely back without a minimum barrier.

11. What size of Effective Sample Size (ESS) for an MCMC trace indicates a weak, poorly-mixed analysis that requires an expanded run length?

- A. An ESS value less than 100
- B. An ESS value exactly at 200
- C. An ESS value tracking near 500
- D. An ESS value exceeding 1000

12. Which tree-building method provides a rooted, ultrametric tree directly by assuming a strict, uniform molecular clock across all lines?

- A. Neighbor-Joining (NJ)
- B. Maximum Likelihood (ML)
- C. Unweighted Pair Group Method with Arithmetic Mean (UPGMA)
- D. Bayesian Inference (BI)

13. What type of evolutionary bias occurs exclusively due to random sampling fluctuations from short alignments, even when the substitution model is 100% correct?

- A. Systematic Bias
- B. Stochastic Bias
- C. Compositional Heterogeneity
- D. Long Branch Attraction

14. What does a gene Concordance Factor (gCF) value of 25% calculated on an internal branch explicitly mean?

- A. The branch has a 25% posterior probability of being correct under a GTR model.
- B. Exactly 25% of individual single-locus gene trees contain that specific branch split.
- C. Only 25% of alignment positions are parsimony-informative.
- D. The molecular clock rate has accelerated by 25% along that lineage.

15. If a fossil is explicitly designated as a 'stem fossil' rather than a 'crown fossil' of a target group, this implies that:

- A. The fossil shares every single synapomorphy present in the living members of the group.
- B. The fossil belongs to an extinct lineage that sits outside the living crown group but is closer to it than to any other living lineage.
- C. The fossil represents the ancestral root sequence of all life on earth.
- D. The fossil cannot be used to establish a hard minimum age constraint on the node.

16. The fundamental mathematical property of a Markov chain substitution model where the future state depends solely on the current state is known as:

- A. Time reversibility.
- B. The Markov property.
- C. Rate homogeneity.
- D. Stationary distribution.

17. Which baseline nucleotide substitution model fixes equilibrium base frequencies at exactly 25% each and uses a single parameter for all mutation rates?

- A. Kimura 2-Parameter (K80)
- B. Felsenstein 1981 (F81)
- C. Jukes-Cantor (JC69)
- D. General Time Reversible (GTR)

18. Which tree topology test assigns relativeSupport probabilities summing to 1 across competing trees rather than using classical alpha p-values to reject models?

- A. Approximately Unbiased (AU) test
- B. Shimodaira-Hasegawa (SH) test
- C. Expected Likelihood Weights (ELW) test
- D. Kishino-Hasegawa (KH) test

19. The continuous trait evolution model that simulates a completely undirected passive random walk is the _____ model, while the model incorporating stabilizing selection parameters pulling traits towards a specific peak is the _____ model.

20. A primary structural advantage of executing an Affine gap cost scoring method over a simple Linear gap cost scoring method is that:

- A. It accelerates alignment computations by removing all insertion blocks.
- B. It accurately mirrors biology by charging a high cost to open a gap but a low cost to extend it, tracking single long insertion/deletion mutations cleanly.
- C. It forces sequences to maintain equal lengths without using any gap penalties.
- D. It functions independently of transition/transversion biases.

21. Non-parametric bootstrapping evaluates branch support by resampling individual alignment columns _____ replacement, whereas Jackknife methods generate pseudo-replicates by deleting a fixed fraction of data columns _____ replacement.

22. In a codon-based multiple sequence alignment structure, all insertions and deletions must occur in strict lengths that are multiples of _____ nucleotides to prevent causing highly disruptive _____ mutations.

23. When independent Bayesian MCMC runs converge on completely distinct likelihood plateaus and topologies, this directly diagnoses:

- A. Excellent chain mixing across the global landscape.
- B. A severe failure of mixing where chains are trapped within localized sub-optimal peaks.
- C. The successful elimination of across-site rate heterogeneity.
- D. The action of strong positive selection on the core dataset.

24. The baseline framework of the Goldman-Yang 1994 (GY94) codon substitution model parameterizes evolutionary pressure in terms of the ratio designated as _____ .

25. If a sequence dataset experiences intense compositional bias where unrelated taxa independently evolve matching GC contents, standard tree reconstruction methods can be severely misled by an artifact called _____ .

26. To rapidly calculate local branch support values in massive trees, the approximate LRT (aLRT) algorithm saves computation time by evaluating only the alternative _____ configuration around the branch of interest.

27. Which of the following amino acid substitution matrices optimized parameter estimations specifically for transmembrane protein structures rather than soluble globular proteins?

- A. Dayhoff's PAM250 matrix
- B. The specialized JTT membrane matrix
- C. WAG matrix
- D. BLOSUM62 matrix

A SINGLE OPEN-ENDED QUESTION WORTH 5 POINTS, ANSWER IN MAX. THREE SENTENCES:

Define what is meant by the concept of heterotachy in sequence evolution, and explicitly explain how failing to incorporate it into a Maximum Likelihood model impacts tree reconstruction accuracy.
